1.How many flip-flops are required to make a MOD-32 binary counter?

A. 3 **B.** 45

C. 5 **D.** 6

2.

Using four cascaded counters with a total of 16 bits, how many states must be deleted to achieve a modulus of 50,000?

A. 50,000 **B.** 65,536

C. 25,536 **D.** 15,536

3.

A MOD-16 ripple counter is holding the count 1001_2. What will the count be after 31 clock pulses?

A. 1000_2 **B.** 1010_2

C. 1011_2 **D.** 1101_2

4.

The terminal count of a modulus-11 binary counter is _____.

A. 1010 **B.** 1000

C. 1001 **D.** 1100

5.

List which pins need to be connected together on a 7493 to make a MOD-12 counter.

A. 12 to 1, 11 to 3, 9 to 2

B. 12 to 1, 11 to 3, 12 to 2

C. 12 to 1, 11 to 3, 8 to 2

D. 12 to 1, 11 to 3, 1 to 2

6.

How can a digital one-shot be implemented using HDL?

A. By using a resistor and a capacitor

B. By applying the concept of a counter

C. By using a library function

D. By applying a level trigger

7. Integrated-circuit counter chips are used in numerous applications including:

A. timing operations, counting operations, sequencing, and frequency multiplication

B. timing operations, counting operations, sequencing, and frequency division

C. timing operations, decoding operations, sequencing, and frequency multiplication

D. data generation, counting operations, sequencing, and frequency multiplication

8. Synchronous construction reduces the delay time of a counter to the delay of:

A. all flip-flops and gates

B. all flip-flops and gates after a 3 count

C. a single gate

D. a single flip-flop and a gate

9. Synchronous counters eliminate the delay problems encountered with asynchronous counters because the:

A. input clock pulses are applied only to the first and last stages

B. input clock pulses are applied only to the last stage

C. input clock pulses are not used to activate any of the counter stages

D. input clock pulses are applied simultaneously to each stage

10. What is the difference between combinational logic and sequential logic?

A. Combinational circuits are not triggered by timing pulses, sequential circuits are triggered by timing pulses.

B. Combinational and sequential circuits are both triggered by timing pulses.

C. Neither circuit is triggered by timing pulses.

11.

What is the difference between a 7490 and a 7492?

A. 7490 is a MOD-12, 7492 is a MOD-10

B. 7490 is a MOD-12, 7492 is a MOD-16

C. 7490 is a MOD-16, 7492 is a MOD-10

D. 7490 is a MOD-10, 7492 is a MOD-12

13.

When two counters are cascaded, the overall MOD number is equal to the _____ of their individual MOD numbers.

A. product **B.** sum

C. log **D.** reciprocal

14.

A MOD-12 and a MOD-10 counter are cascaded. Determine the output frequency if the input clock frequency is 60 MHz.

A. 500 kHz

B. 1,500 kHz

C. 6 MHz

D. 5 MHz

15.

Which segments of a seven-segment display would be required to be active to display the decimal digit 2?

A. *a*, *b*, *d*, *e*, and *g*

B. *a*, *b*, *c*, *d*, and *g*

C. *a*, *c*, *d*, *f*, and *g*

D. *a*, *b*, *c*, *d*, *e*, and *f*

16.

How many AND gates would be required to completely decode ALL the states of a MOD-64 counter, and how many inputs must each AND gate have?

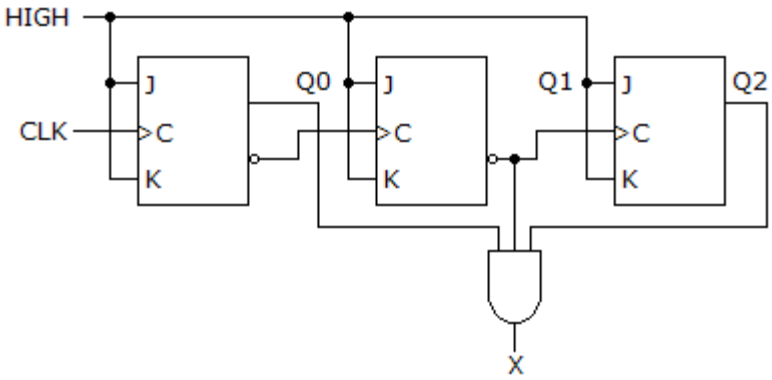
A. 128 gates, 6 inputs to each gate

B. 64 gates, 5 inputs to each gate

C. 64 gates, 6 inputs to each gate

D. 128 gates, 5 inputs to each gate

17. What decimal value is required to produce an output at "X" ?



A. 1

B. 1 or 4

C. 2

D. 5

18. A BCD counter is a _____.

A. binary counter

B. full-modulus counter

C. decade counter

D. divide-by-10 counter

20. How many flip-flops are required to construct a decade counter?

A. 10 **B.** 8

C. 5 D. 4

21. The terminal count of a typical modulus-10 binary counter is _____.

A. 0000 B. 1010

C. 1001 D. 1111

22. A seven-segment, common-anode LED display is designed for:

A. all cathodes to be wired together

B. one common LED

C. a HIGH to turn off each segment

D. disorientation of segment modules

23. To operate correctly, starting a ring counter requires:

A. clearing one flip-flop and presetting all the others.

B. clearing all the flip-flops.

C. presetting one flip-flop and clearing all the others.

D. presetting all the flip-flops.

24. The process of designing a synchronous counter that will count in a nonbinary manner is primarily based on:

A. external logic circuits that decode the various states of the counter to apply the correct logic levels to the *J-K* inputs

B. modifying BCD counters to change states on every second input clock pulse

C. modifying asynchronous counters to change states on every second input clock pulse

D. elimination of the counter stages and the addition of combinational logic circuits to produce the desired counts

25. Select the response that best describes the use of the Master Reset on typical 4-bit binary counters.

A. When MR_1 and MR_2 are both HIGH, all Qs will be reset to zero.

B. When MR_1 and MR_2 are both HIGH, all Qs will be reset to one.

C. MR_1 and MR_2 are provided to synchronously reset all four flip-flops.

D. To enable the count mode, MR_1 and MR_2 must be held LOW.

26. For a multistage counter to be truly synchronous, the _____ of each stage must be connected to _____.

A. C_p, the same clock input line

B. CE, the same clock input line

C. $\overline{PL}$, the terminal count output

D. $\overline{RC}$, both clock input lines

27. Which of the following is an invalid output state for an 8421 BCD counter?

A. 1110 **B.** 0000

C. 0010 **D.** 0001

28. How many different states does a 3-bit asynchronous counter have?

A. 2 **B.** 4

C. 8 **D.** 16

29. A 5-bit asynchronous binary counter is made up of five flip-flops, each with a 12 ns propagation delay. The total propagation delay ($t_{p(tot)}$) is _____.

A. 12 ms

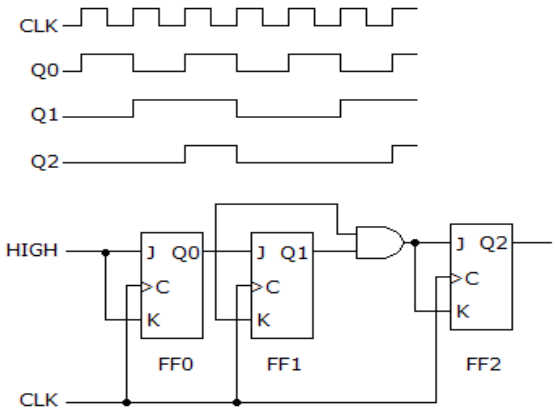
B. 24 ns

C. 48 ns

D. 60 ns

30.

A four-channel scope is used to check the counter in the figure given below. Are the displayed waveforms correct?



A. Yes **B.** No

32.

One of the major drawbacks to the use of asynchronous counters is:

A. low-frequency applications are limited because of internal propagation delays

B. high-frequency applications are limited because of internal propagation delays

C. asynchronous counters do not have major drawbacks and are suitable for use in high- and low-frequency counting applications

D. asynchronous counters do not have propagation delays and this limits their use in high-frequency applications

33.

Once an up-/down-counter begins its count sequence, it cannot be reversed.

A. True **B.** False

34.

Three cascaded modulus-5 counters have an overall modulus of _____.

A. 5 **B.** 25

C. 125 **D.** 500

35.

An asynchronous 4-bit binary down counter changes from count 2 to count 3. How many transitional states are required?

A. None **B.** One

C. Two D. Fifteen

36. The final output of a modulus-8 counter occurs one time for every _____.

A. 8 clock pulses

B. 16 clock pulses

C. 24 clock pulses

D. 32 clock pulses

37. A 4-bit up/down binary counter is in the DOWN mode and in the 1100 state. To what state does the counter go on the next clock pulse?

A. 1101 B. 1011

C. 1111 D. 0000

38. A 4-bit ripple counter consists of flip-flops, which each have a propagation delay from clock to Q output of 15 ns. For the counter to recycle from 1111 to 0000, it takes a total of _____.

A. 15 ns

B. 30 ns

C. 45 ns

D. 60 ns

39. The terminal count of a 3-bit binary counter in the DOWN mode is _____.

A. 000 B. 111

C. 101 D. 010

40. The hexadecimal equivalent of 15,536 is _____.

A. 3CB0 B. 3C66

C. 63C0 D. 6300

41. In an HDL ring counter, many invalid states are included in the programming by:

A. using a case statement.

B. using an elsif statement.

C. including them under others.

D. the *ser_in* line.

42.

In a VHDL retriggerable edge-triggered one-shot, which condition will not exist when a clock edge occurs?

A. A trigger edge has occurred and we must load the counter.

B. The counter is zero and we need to keep it at zero.

C. The shift register is reset.

D. The counter is not zero and we need to count down by one.

44.

Synchronous (parallel) counters eliminate the delay problems encountered with asynchronous (ripple) counters because the:

A. input clock pulses are applied only to the first and last stages.

B. input clock pulses are applied only to the last stage.

C. input clock pulses are applied simultaneously to each stage.

D. input clock pulses are not used to activate any of the counter stages.

45.

List the state of each output pin of a 7447 if $RBI = 0$, $LT = 1$, $A_0 = 1$, $A_1 = 0$, $A_2 = 0$, and $A_3 = 1$.

A. $RBO = 0, a = 0, b = 0, c = 0, d = 1, e = 1, f = 0, g = 0$

B. $RBO = 1, a = 0, b = 0, c = 0, d = 1, e = 1, f = 0, g = 0$

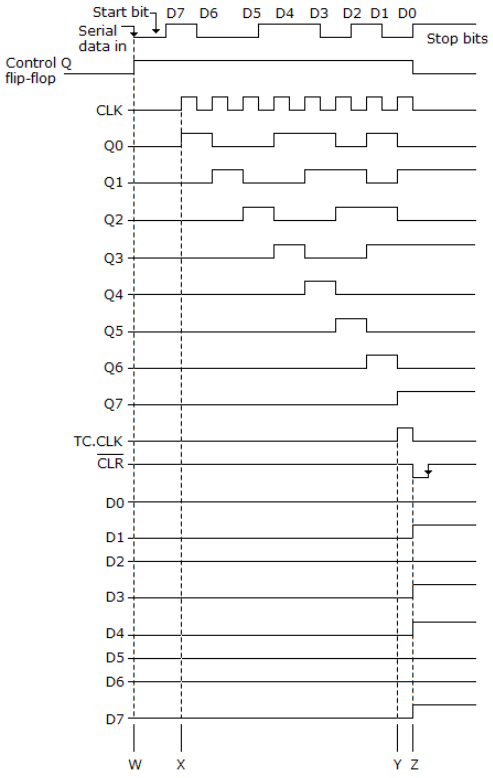
C. $RBO = 0, a = 0, b = 0, c = 0, d = 0, e = 1, f = 0, g = 0$

D. $RBO = 1, a = 0, b = 0, c = 0, d = 0, e = 1, f = 0, g = 0$

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46.

Referring to the given figure, what causes the Control FF to reset after D7?



A. Once the data cycle is initiated by the Start bit, the one-shot produces an output pulse equal to the duration of the eight data bits. Once the eight data bits have been transferred to the data input register, the falling edge of the one-shot pulse resets the Control FF to start the sequence all over again.

B. After counting the eight data bits, the divide-by-8 counter produces an output on its active-LOW CLR line to reset the Control FF.

C. After counting eight clock pulses equivalent to eight data periods, the terminal count of the divide-by-8 counter and the clock trigger the one-shot, which in turn resets the Control FF and divide-by-8 circuits to begin the sequence all over again. Simultaneously the data is transferred through the output register.

D. When the data output register is full, it produces an output on its *C* terminal that triggers the one-shot, which in turn resets the Control FF.

48.

Three cascaded decade counters will divide the input frequency by _____.

A. 10 **B.** 20

C. 100 D. 1,000

49. A counter with a modulus of 16 acts as a _____.

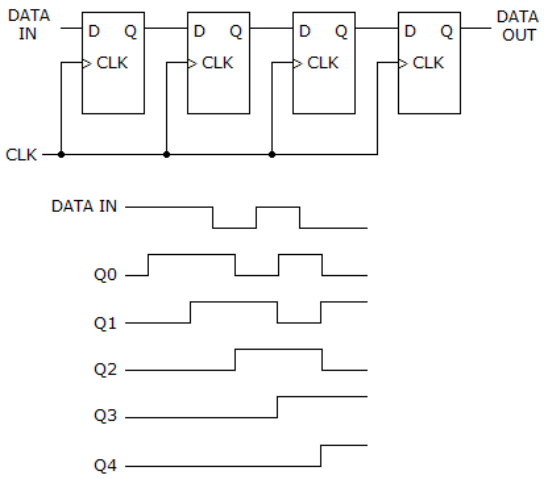
A. divide-by-8 counter

B. divide-by-16 counter

C. divide-by-32 counter

D. divide-by-64 counter

50. How many data bits can be stored in the register shown below?



A. 5 B. 32

C. 31 D. 4

51. What is the difference between a 7490 and a 7493?

A. 7490 is a MOD-10, 7493 is a MOD-16

B. 7490 is a MOD-16, 7493 is a MOD-10

C. 7490 is a MOD-12, 7493 is a MOD-16

D. 7490 is a MOD-10, 7493 is a MOD-12

52. A ripple counter's speed is limited by the propagation delay of:

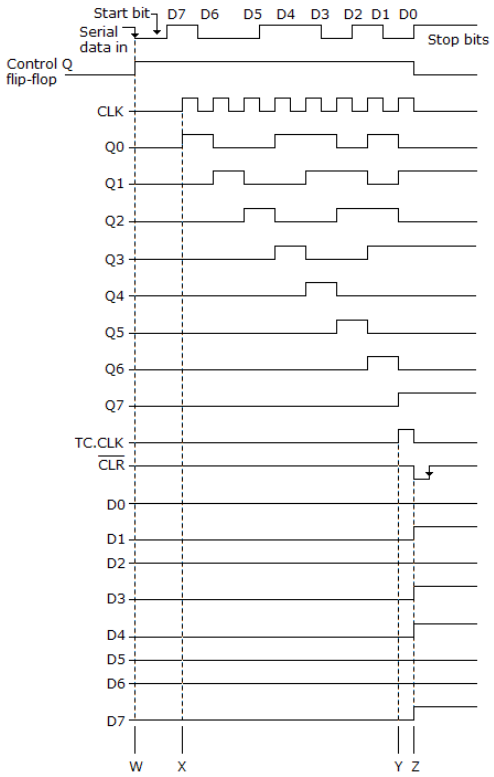
A. each flip-flop

B. all flip-flops and gates

C. the flip-flops only with gates

D. only circuit gates

53. Referring to the given figure, at which point is the serial data transferred to the parallel output?



A. W **B.** X

C. Y **D.** Z

54. A 4-bit counter has a maximum modulus of _____.

A. 3 **B.** 6

C. 8 **D.** 16

55. Which of the following statements best describes the operation of a synchronous up-/down-counter?

A. The counter can count in either direction, but must continue in that direction once started.

B. The counter can be reversed, but must be reset before counting in the other

direction.

C. In general, the counter can be reversed at any point in its counting sequence.

D. The count sequence cannot be reversed, once it has begun, without first resetting the counter to zero.

56. The parallel outputs of a counter circuit represent the:

A. parallel data word

B. clock frequency

C. counter modulus

D. clock count

57. Any divide-by-*N* counter can be formed by using external gating to _____ at a predetermined number.

A. HIGH

B. reset

C. LOW

D. preset

59. How many natural states will there be in a 4-bit ripple counter?

A. 4

B. 8

C. 16

D. 32

60. List which pins need to be connected together on a 7492 to make a MOD-12 counter.

A. 1 to 12, 11 to 6, 9 to 7

B. 1 to 12, 12 to 6, 11 to 7

C. 1 to 12, 9 to 6, 8 to 7

D. 1 to 12

61. A principle regarding most display decoders is that when the correct input is present, the

related output will switch:

A. HIGH

B. to high impedance

C. to an open

D. LOW

62. A modulus-10 counter must have _____.

A. 10 flip-flops

B. flip-flops

C. 2 flip-flops

D. synchronous clocking

63. For a one-shot application, how can HDL code be used to make a circuit respond once to each positive transition on its trigger input?

A. By using a counter

B. By using an active clock

C. By using an immediate reload

D. By using edge trapping

64. Which is not an example of a truncated modulus?

A. 8 **B.** 9

C. 11 **D.** 15

65. Four cascaded modulus-10 counters have an overall modulus of _____.

A. 10 **B.** 100

C. 1,000 **D.** 10,000

66. What is the maximum delay that can occur if four flip-flops are connected as a ripple

counter and each flip-flop has propagation delays of t_{PHL} = 22 ns and t_{PLH} = 15 ns?

A. 15 ns

B. 22 ns

C. 60 ns

D. 88 ns

67.

Which of the following statements are true?

A. Asynchronous events do not occur at the same time.

B. Asynchronous events are controlled by a clock.

C. Synchronous events do not need a clock to control them.

D. Only asynchronous events need a control clock.

69.

Which of the following groups of logic devices would be the minimum required for a MOD-64 synchronous counter?

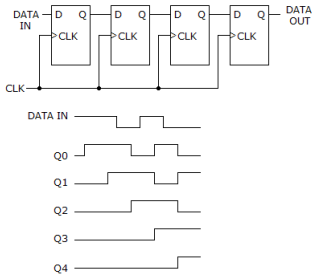
A. Five flip-flops, three AND gates

B. Seven flip-flops, five AND gates

C. Four flip-flops, ten AND gates

D. Six flip-flops, four AND gates

70. The circuit given below fails to produce data output. The individual flip-flops are checked with a logic probe and pulser, and each checks OK. What could be causing the problem?



A. The data output line may be grounded.

B. One of the clock input lines may be open.

C. One of the interconnect lines between two stages may have a solder bridge to ground.

D. One of the flip-flops may have a solder bridge between its input and V_{cc}.

71. A 22-MHz clock signal is put into a MOD-16 counter. What is the frequency of the *Q*output of each stage of the counter?

A. $Q_1 = 22$ MHz, $Q_2 = 11$ MHz, $Q_3 = 5.5$ MHz, $Q_4 = 2.75$ MHz

B. $Q_1 = 11$ MHz, $Q_2 = 5.5$ MHz, $Q_3 = 2.75$ MHz, $Q_4 = 1.375$ MHz

C. $Q_1 = 11$ MHz, $Q_2 = 11$ MHz, $Q_3 = 11$ MHz, $Q_4 = 11$ MHz

D. $Q_1 = 22$ MHz, $Q_2 = 22$ MHz, $Q_3 = 22$ MHz, $Q_4 = 22$ MHz

72. The designation $\text{UP}/\overline{\text{DOWN}}$ means that the _____.

A. up count is active-HIGH, the down count is active-LOW

B. up count is active-LOW, the down count is active-HIGH

C. up and down counts are both active-LOW

D. up and down counts are both active-HIGH

74. Why can a synchronous counter operate at a higher frequency than a ripple counter?

A. The flip-flops change one after the other.

B. The flip-flops change at the same time.

C. A synchronous counter cannot operate at higher frequencies.

D. A ripple counter is faster.

75.

A multiplexed display being driven by a logic circuit:

A. accepts data inputs from one line and passes this data to multiple output lines

B. accepts data inputs from several lines and allows one of them at a time to pass to the output

C. accepts data inputs from multiple lines and passes this data to multiple output lines

D. accepts data inputs from several lines and multiplexes this input data to four BCD lines

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76.

What is meant by parallel load of a counter?

A. Each FF is loaded with data on a separate clock.

B. The counter is cleared.

C. All FFs are preset with data.

77.

Which of the following is an example of a counter with a truncated modulus?

A. 8 **B.** 13

C. 16 **D.** 32

78.

Which of the following is a type of shift register counter?

A. Decade **B.** Binary

C. Ring **D.** BCD

79.

MOD-6 and MOD-12 counters and multiples are most commonly used as:

A. frequency counters

B. multiplexed displays

C. digital clocks

D. power consumption meters

80. Which of the following is an invalid state in an 8421 BCD counter?

A. 0011 **B.** 1001

C. 1000 **D.** 1100

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81. After 10 clock cycles, and assuming that the DATA input had returned to 0 following the storage sequence, what values would be stored in *Q*4, *Q*3, *Q*2, *Q*1, *Q*0 of the register in Figure 7-5?

A. 0,1,0,1,1 **B.** 1,1,0,1,0

C. 1,0,1,0,1 **D.** 0,0,0,0,0

82. How many different states does a 2-bit asynchronous counter have?

A. 1 **B.** 2

C. 4 **D.** 8

83. A 12 MHz clock frequency is applied to a cascaded counter containing a modulus-5 counter, a modulus-8 counter, and a modulus-10 counter. The lowest output frequency possible is _____.

A. 10 kHz

B. 20 kHz

C. 30 kHz

D. 60 kHz